

children at younger ages test hypotheses, detect statistical patterns and draw conclusions while constantly adapting to changes.

"Young children are capable of solving problems that still pose a challenge for computers, such as learning languages and figuring out causal relationships," said study co-author Tom Griffiths, director of Berkeley's Computational Cognitive Science Lab.

"We are hoping to make computers smarter by making them a little more like children. Your computer could be able to discover causal relationships, ranging from simple cases such as recognising that you work more slowly when you haven't had coffee, to complex ones such as identifying which genes cause greater susceptibility to diseases," said Griffiths.

Computers powered by swarms of crabs

It has been discovered that swarms of soldier crabs herded through tunnels can form the AND, OR and NOT logic gates required to make a computer.

When two swarms of crabs collide, they merge and continue in a direction that is the sum of their velocities. A team of researchers at Kobe University, Japan, has applied this behaviour of crabs to computing. They adapted a previous model of unconventional computing—based on colliding billiard balls—to work with swarms of crabs, with 0's and 1's represented by the absence and presence of a swarm, respectively.

Researchers first tried the idea with simulated crab swarms. The OR gate, which simply combines one or two crab swarms into one, worked every time, but the more complicated AND gate, which involves the combined swarm heading down one of three paths, was less reliable.

They then implemented the logic gates using swarms of 40 crabs. The swarms were placed at the entrances of the logic gates and encouraged to move by a looming shadow that fooled them into thinking a predatory bird was overhead. The results closely matched the simulation, suggesting that crab-powered computers could indeed be possible.

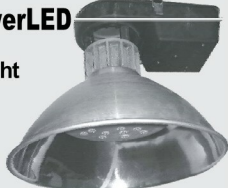
Metal detector that can identify buried bombs

A metal detector developed by Eugene Lavelly of BAE Systems in Massachusetts and his team can tell whether buried objects are unexploded bombs or just trash. This would open up the possibility of governments cleaning the bombs out to be able to reuse the land they are located in.

By using a technique called time-domain electromagnetic induction, a coil sends an electromagnetic pulse 15 metres into the ground. The pulse makes the things it hits reverberate like a struck drum. The team has identified the reverberation signal of a torpedo-shaped metal object with a hollow core—where explosives may lie.

The researchers are now refining the method so that it is accurate enough to meet US standards, which require 99.9 per cent confidence that all bombs have been dug up before a land can be used.

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